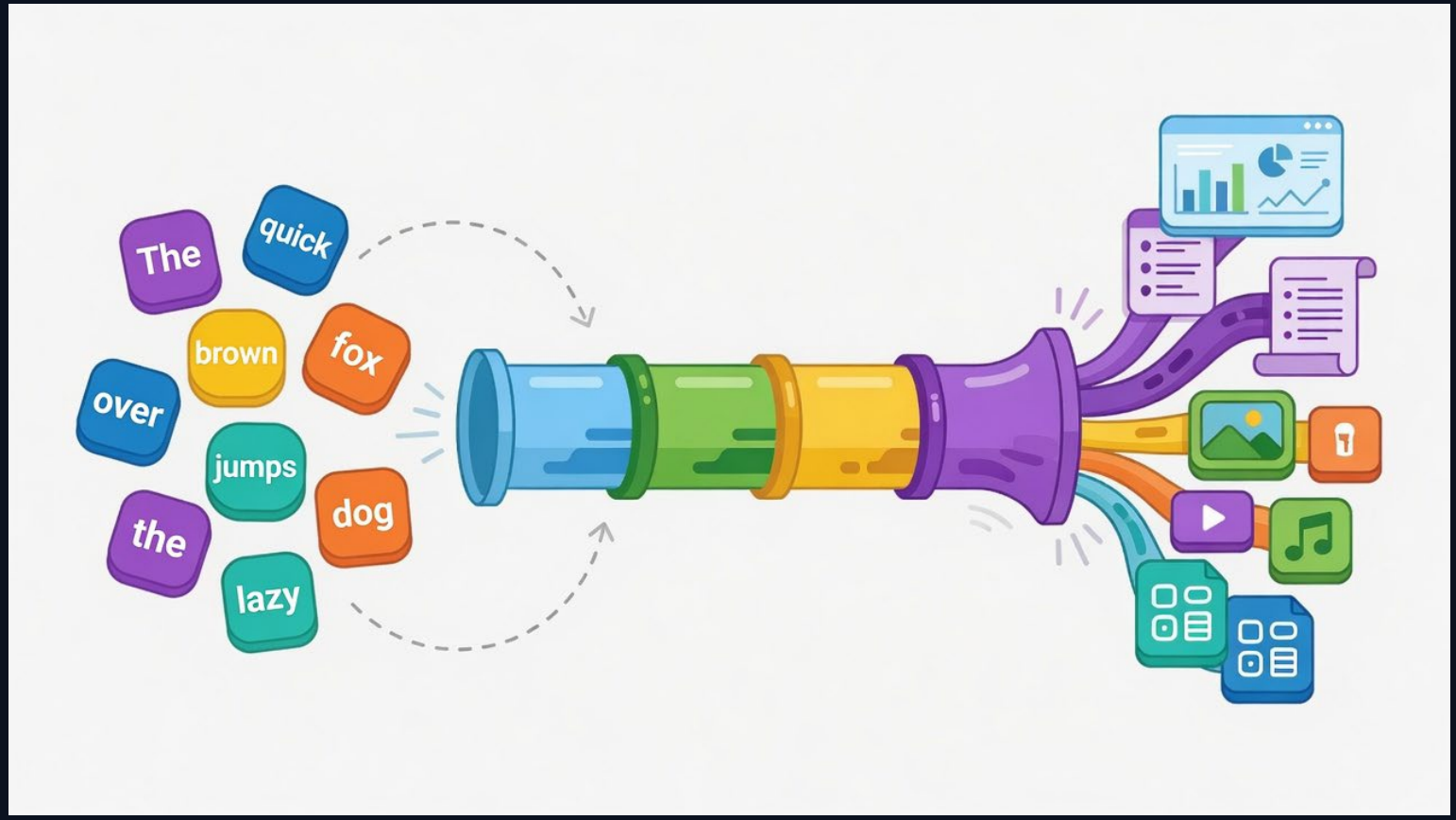
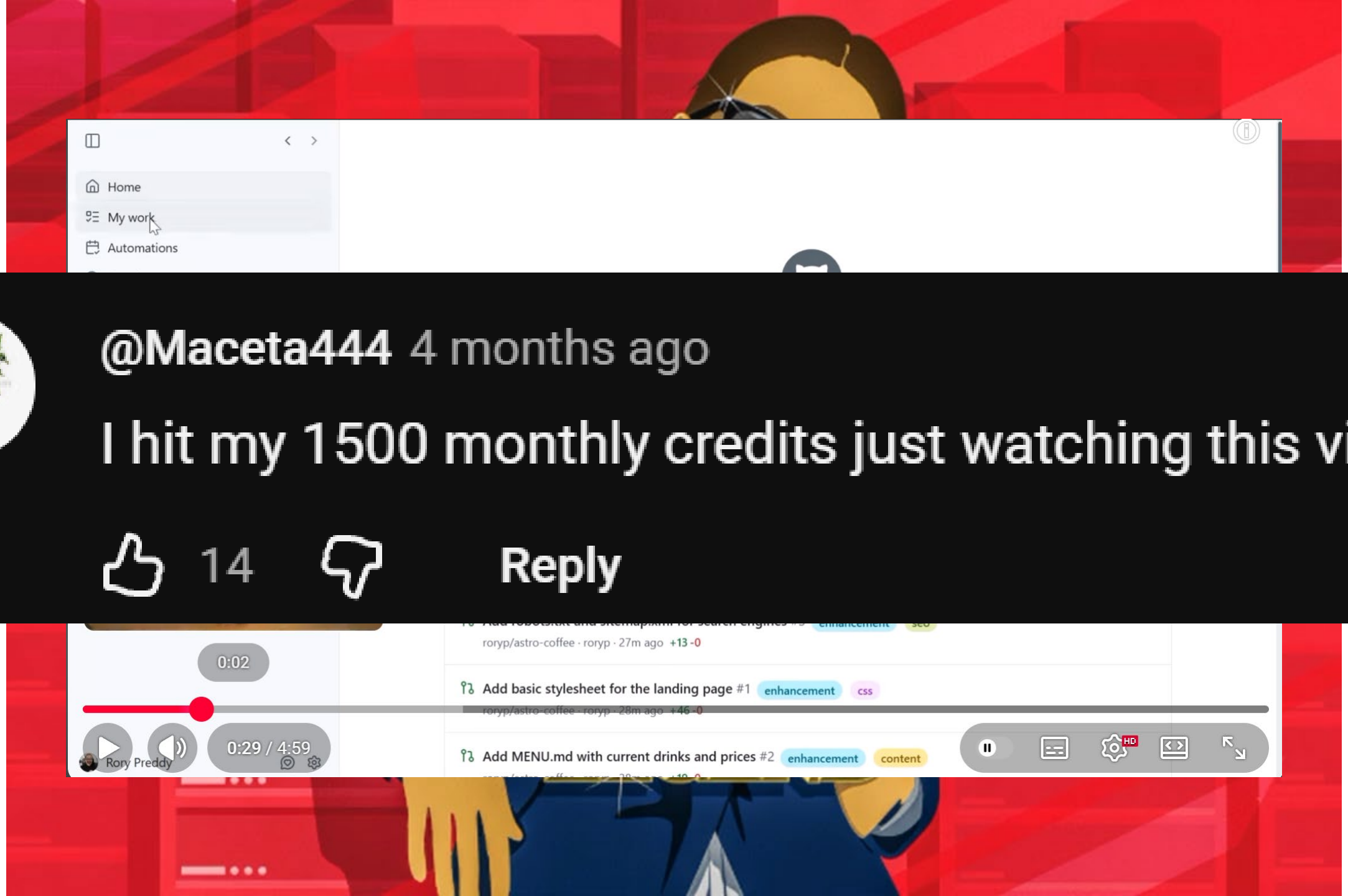
Token Economics



Agenda

- Introduction
- Demo 1 - 4 levers
- Demo 2 - Pricing
- Tracing and Scaling
- Resources


Home
My work
Automations
@Maceta444 4 months ago
I hit my 1500 monthly credits just watching this video
14
Reply
0:02
0:29 / 4:59
Rory Preddy
roryp/astro-coffee · roryp · 27m ago +13 -0
Add basic stylesheet for the landing page #1
enhancement
css
roryp/astro-coffee · roryp · 28m ago +46 -0
Add MENU.md with current drinks and prices #2
enhancement
content
HD

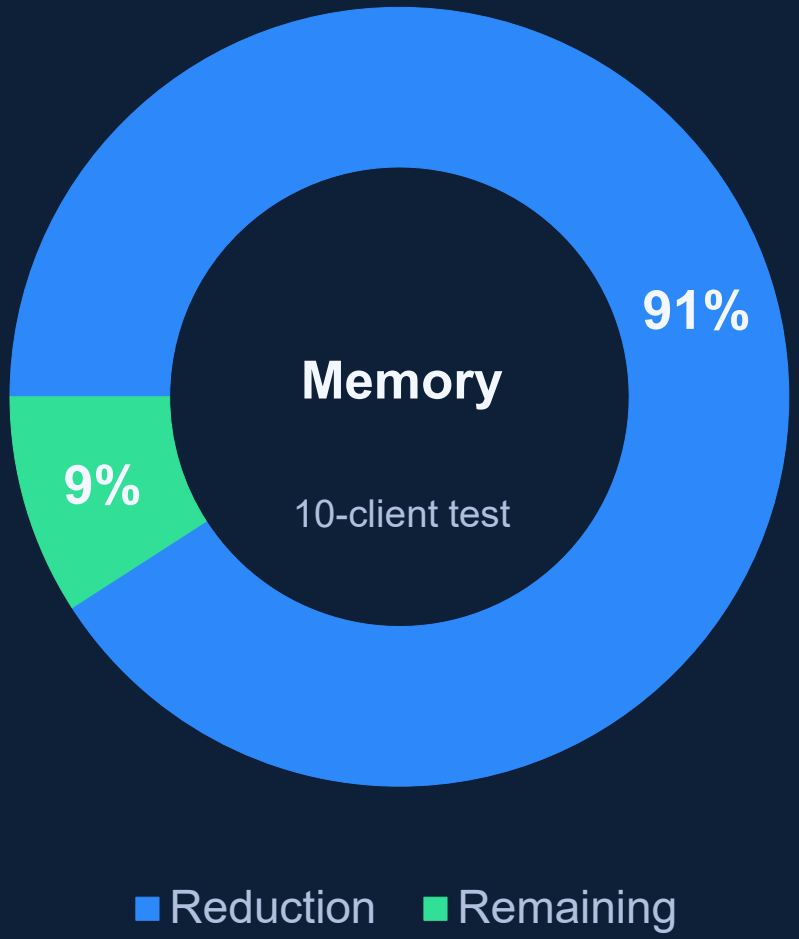
GitHub Copilot Runtime Ported to Rust

A faster, leaner foundation for agents



https://aka.ms/copilot-migration

Copilot runtime performance



91%

less memory

1,383 MB before / 126 MB after

65%

less processor time

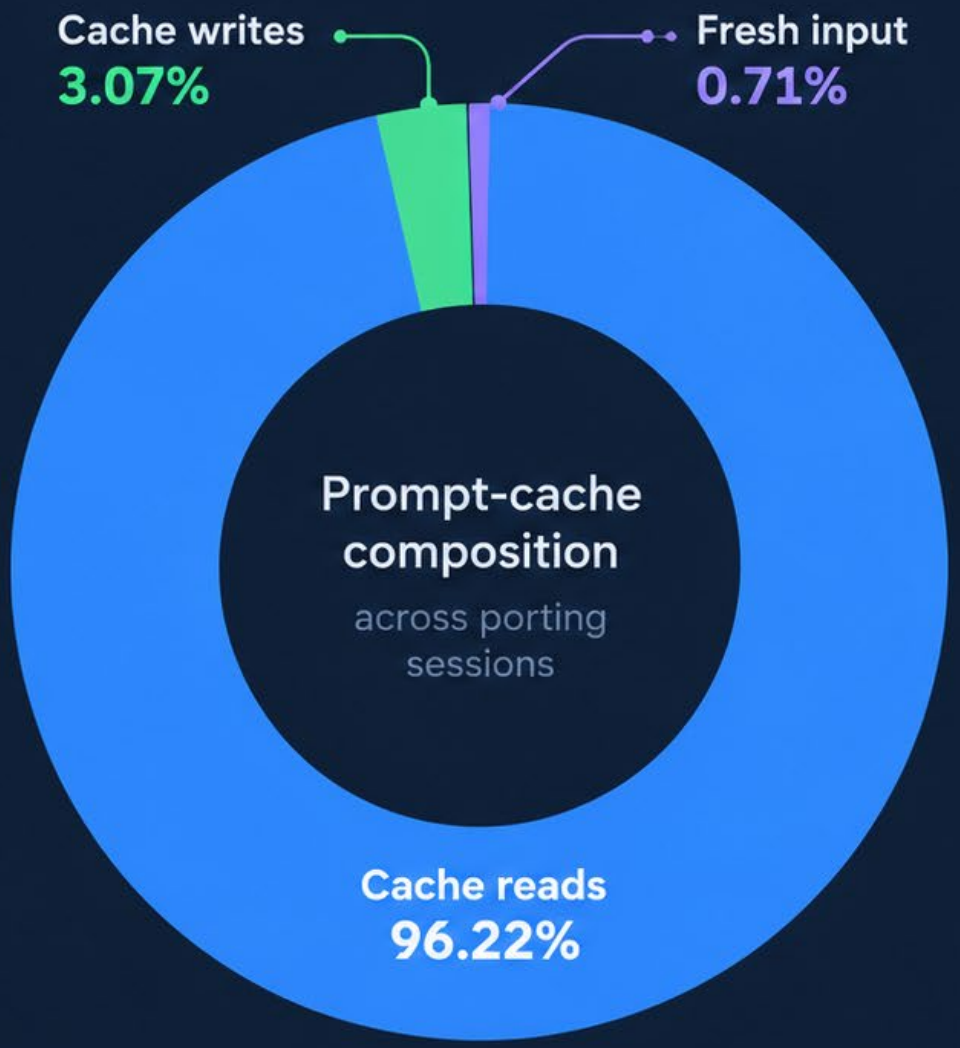
312 s before / ~110 s after

4×

faster session lifecycle

5.25 s before / 1.33 s after

Cost savings



$120K
total spend

$0.14
per line of production Rust

136.3B
total tokens

130.6B cached reads · **4.2B** cache writes · **0.9B** fresh input · **0.6B** output

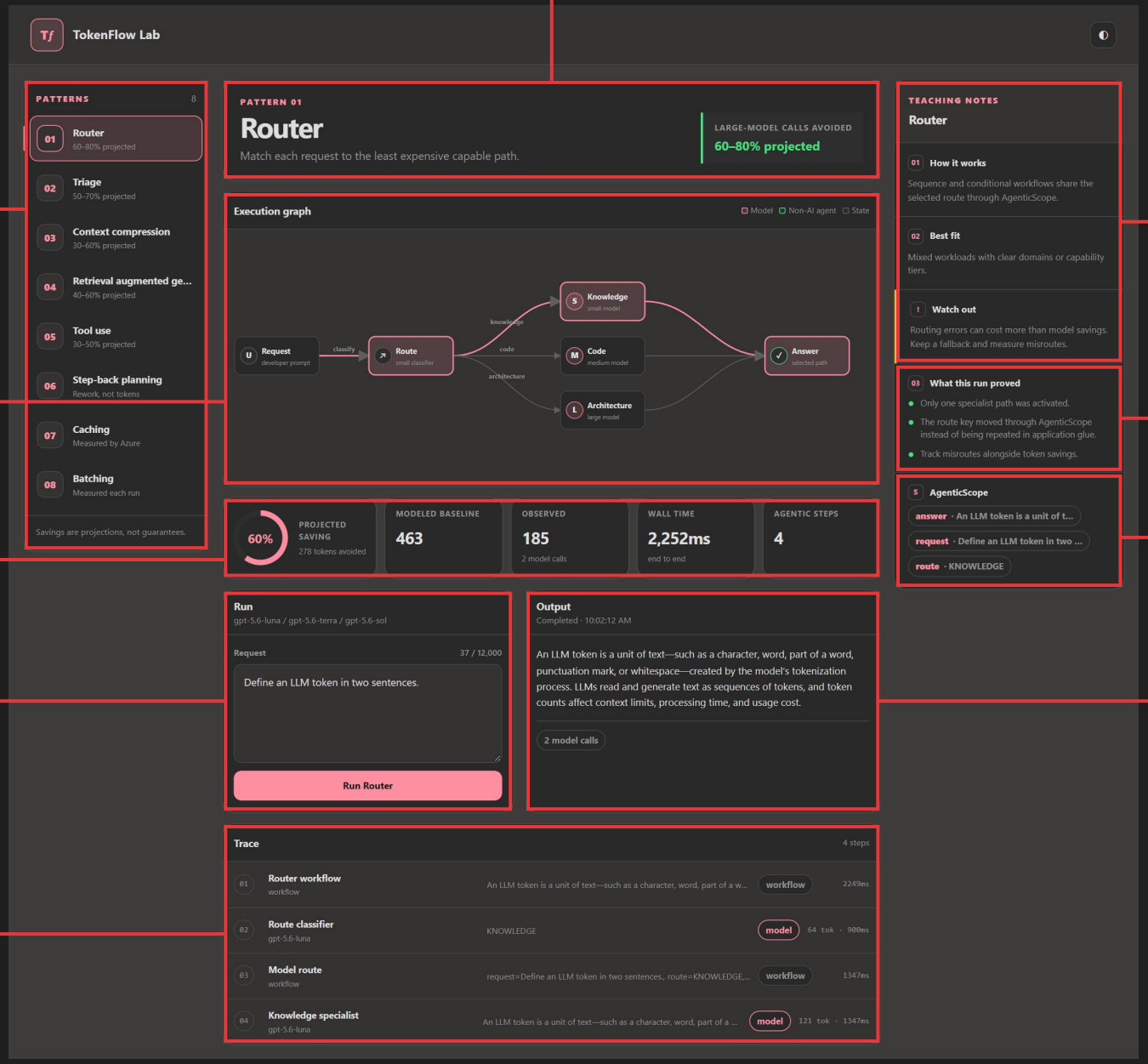
96.22% prompt-cache hit rate — primary savings lever

128 PRs

Agenda

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- Resources

TOKENFLOW LAB



1. Patterns

2. Pattern + goal

3. Execution graph

4. Token metrics

5. Request + run

6. Answer

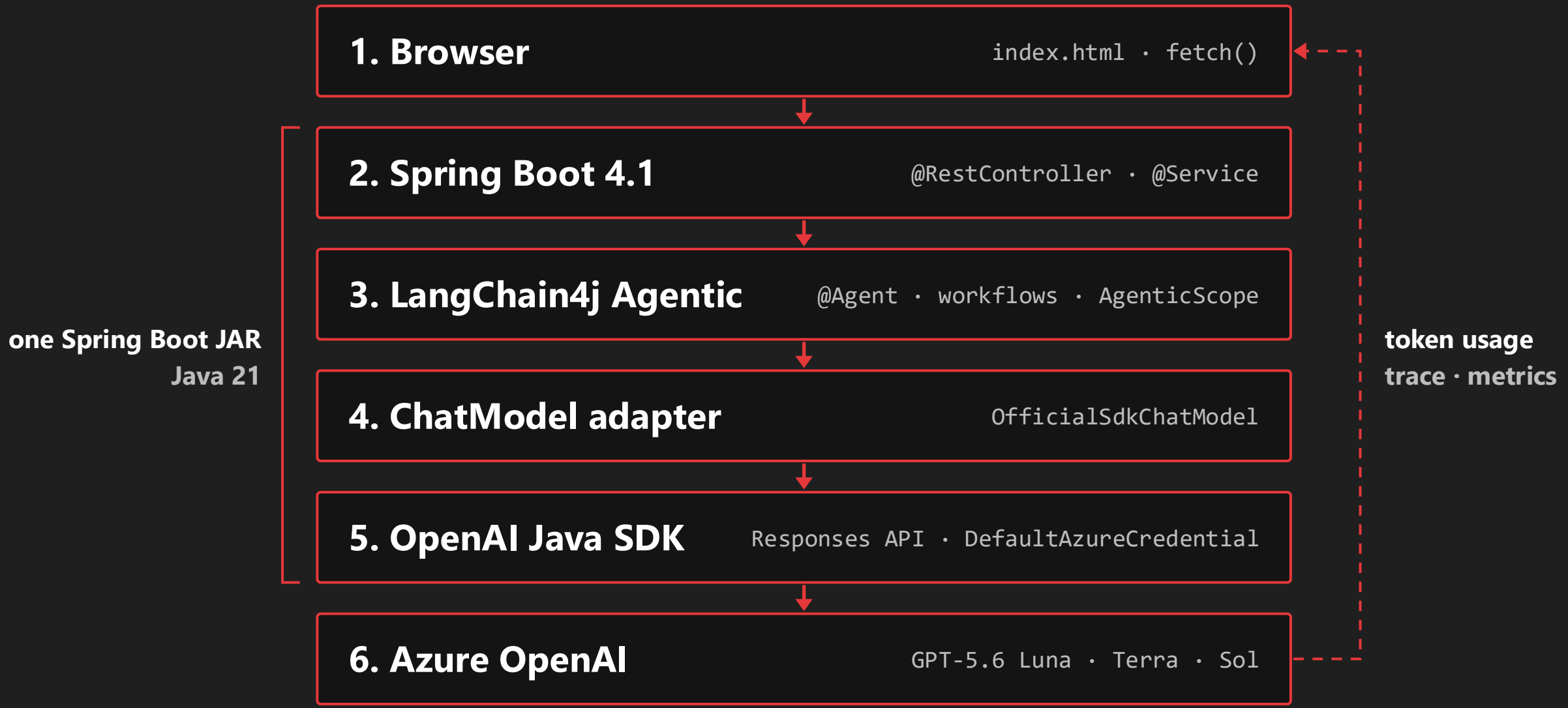
7. Trace

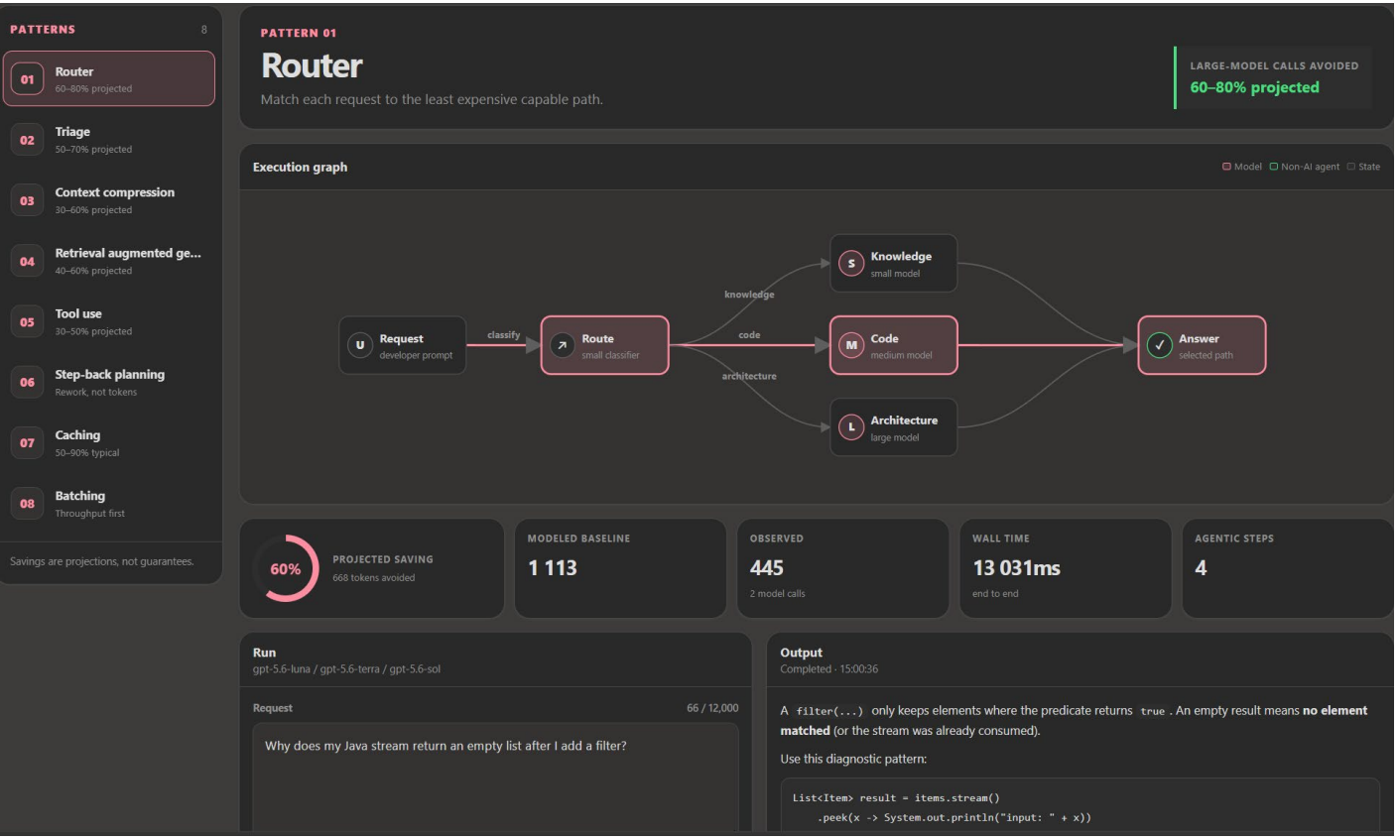
8. Teaching notes

9. What it proved

10. AgenticScope

SPRING + LANGCHAIN4J

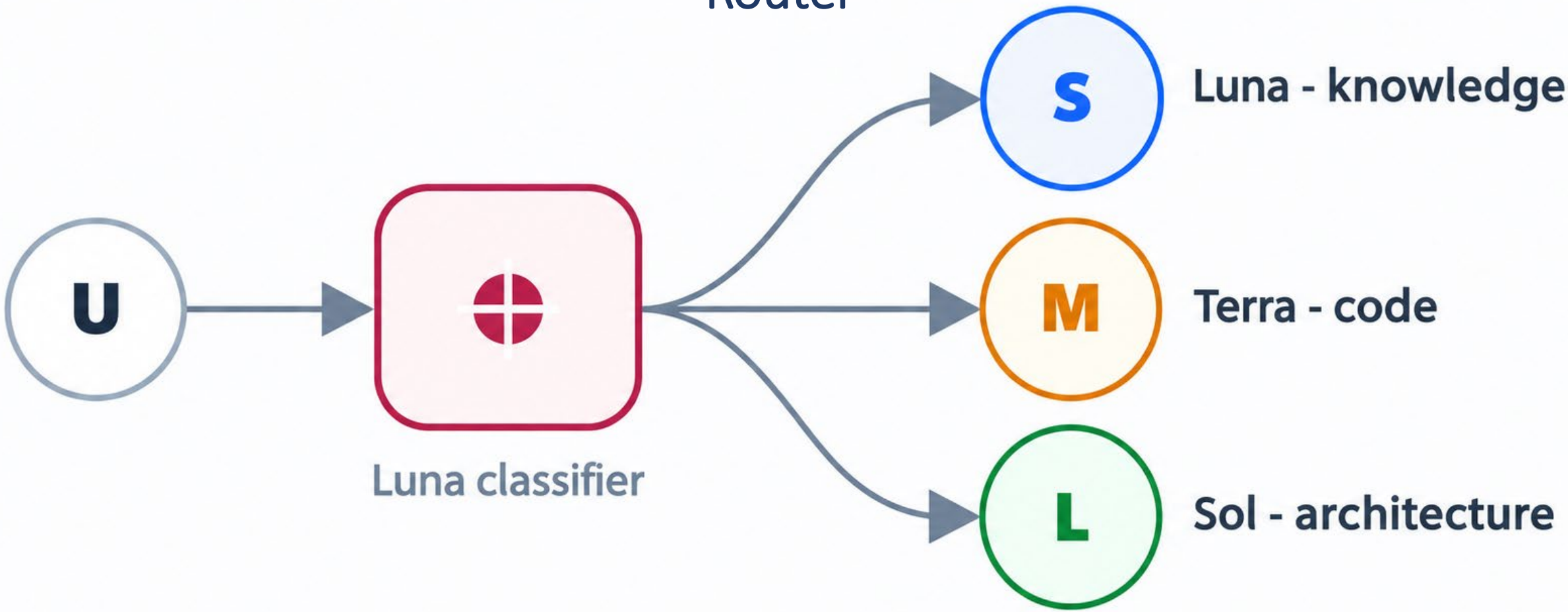




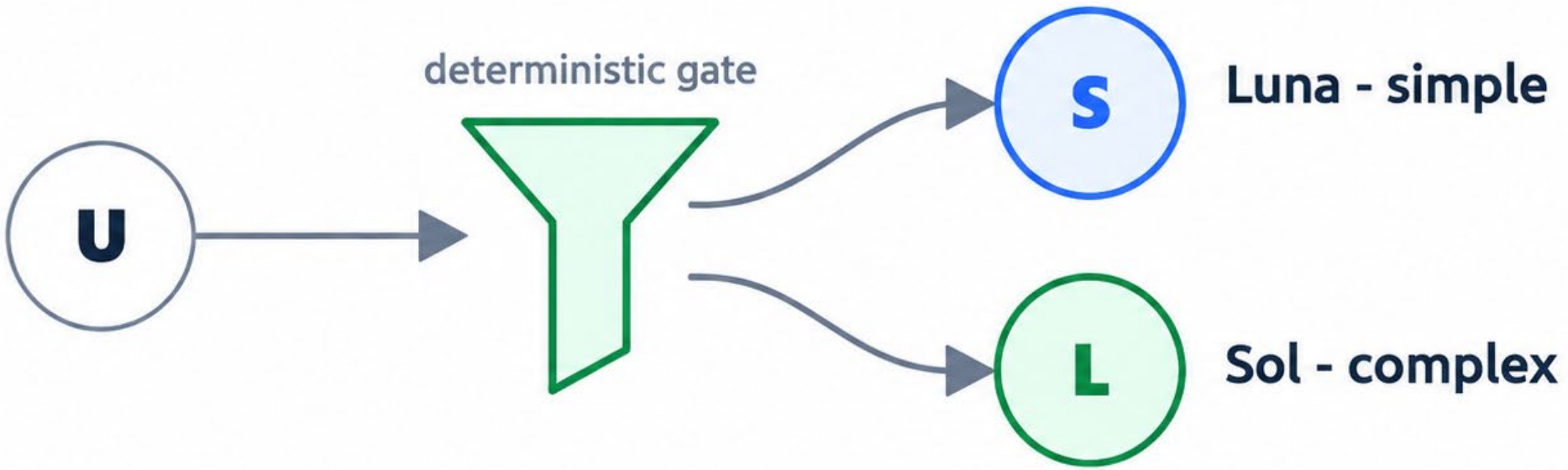
https://aka.ms/costs


MODEL
Router
Triage
PROMPT
Step-back planning
CONTEXT
Compression
RAG
Tool use
Caching
OUTPUT
Batching

Router

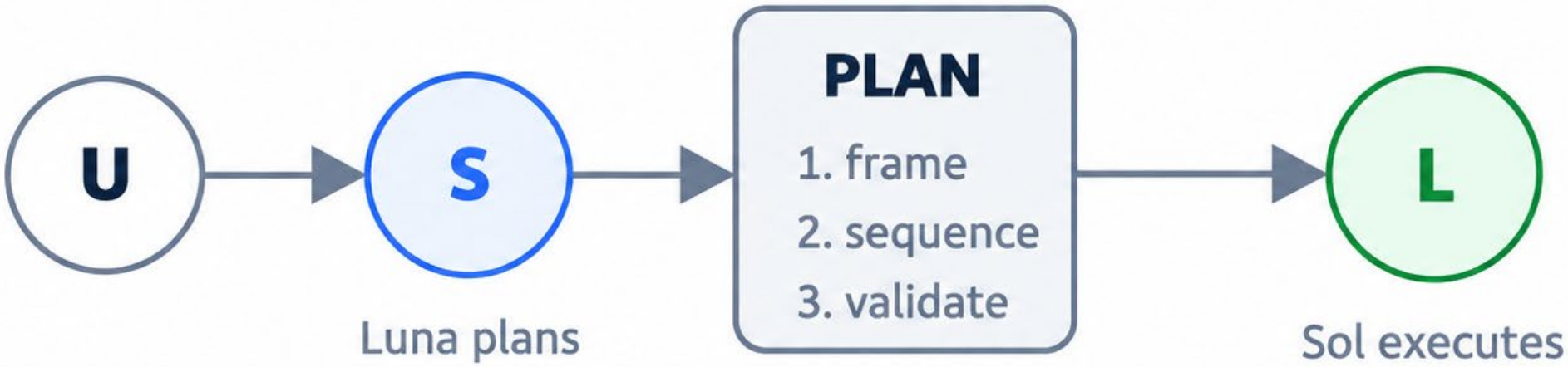

U
Luna classifier
S
Luna - knowledge
M
Terra - code
L
Sol - architecture

Triage



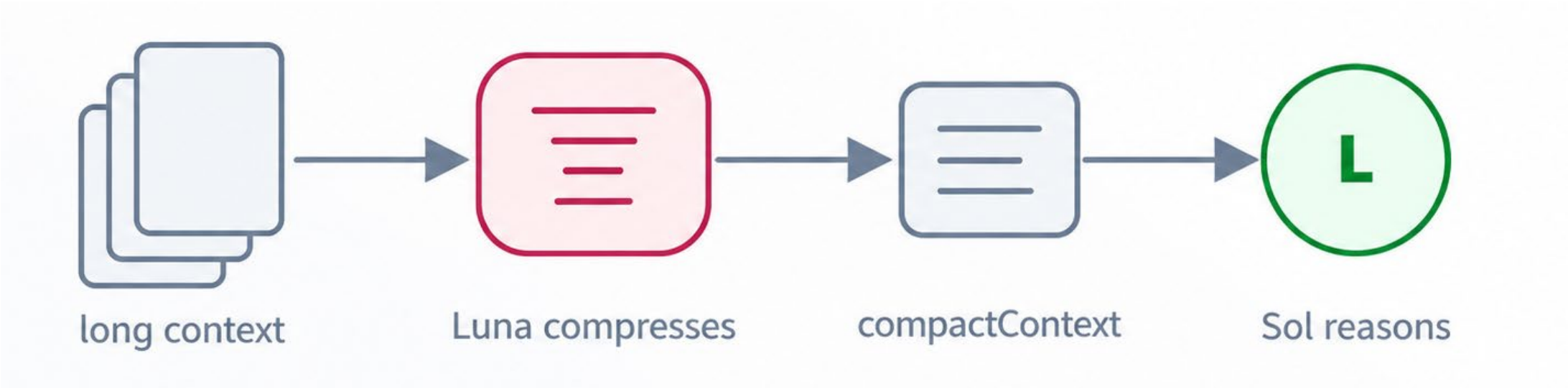

MODEL
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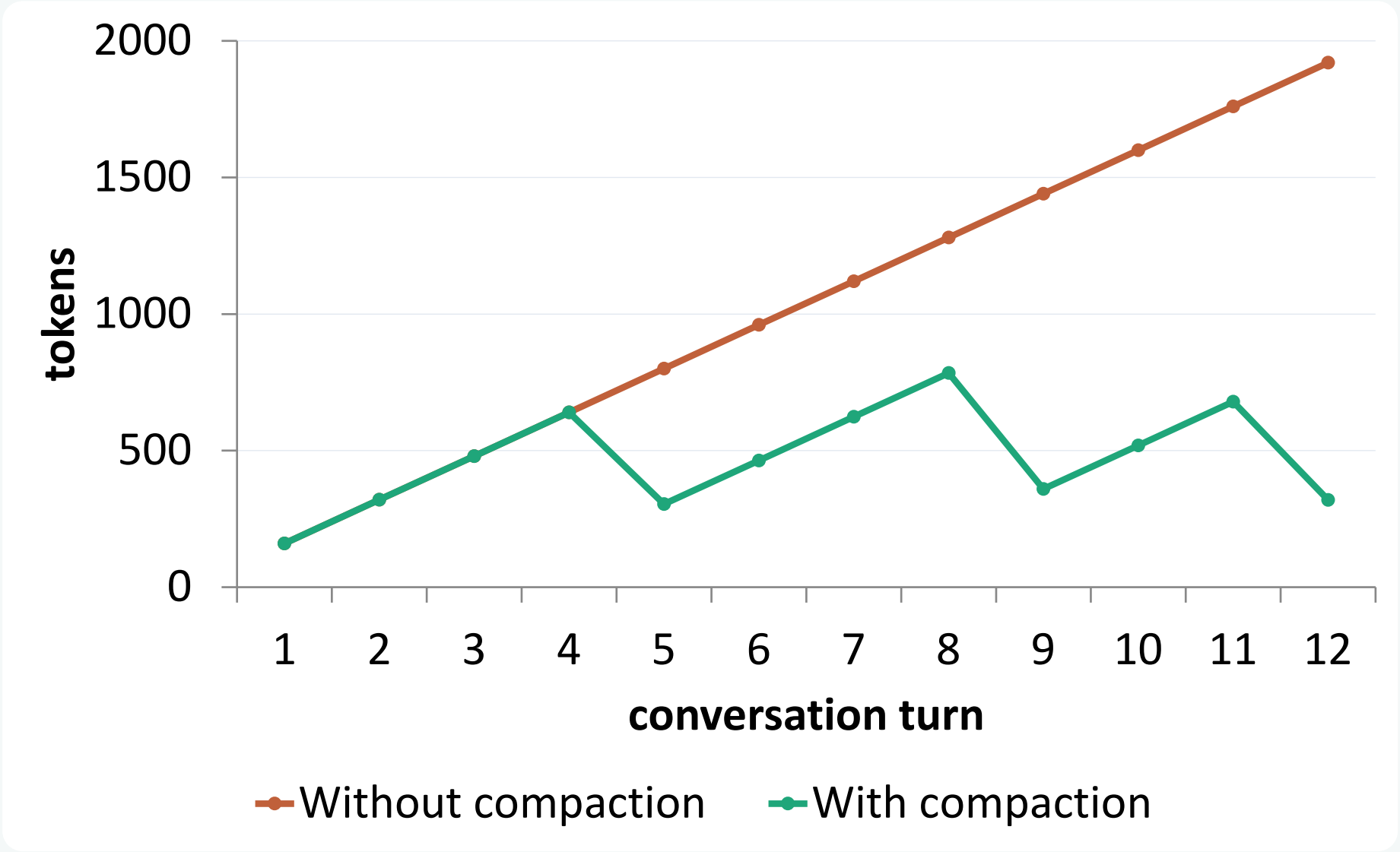
Step-back planning



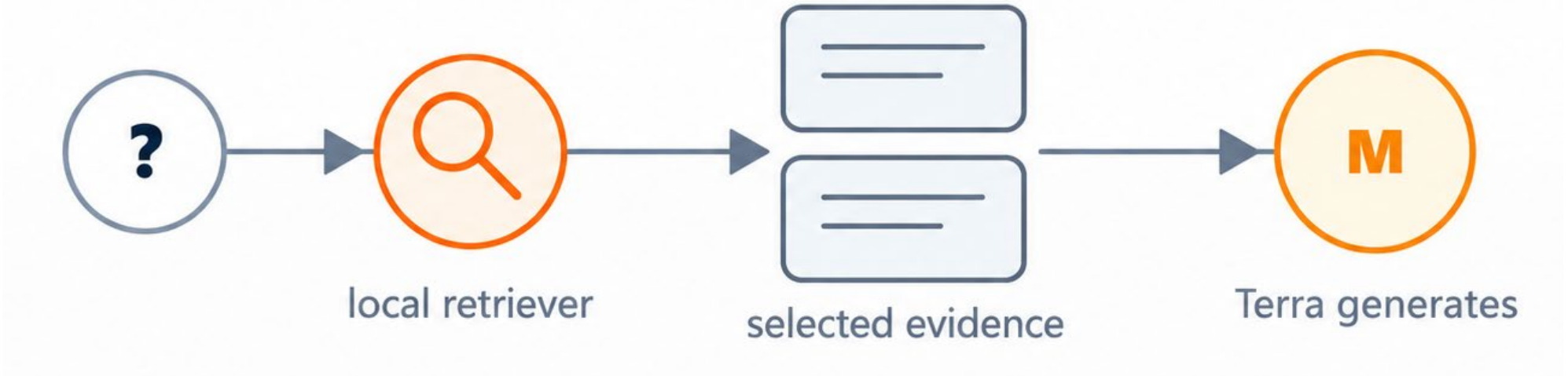

MODEL
PROMPT
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Compression

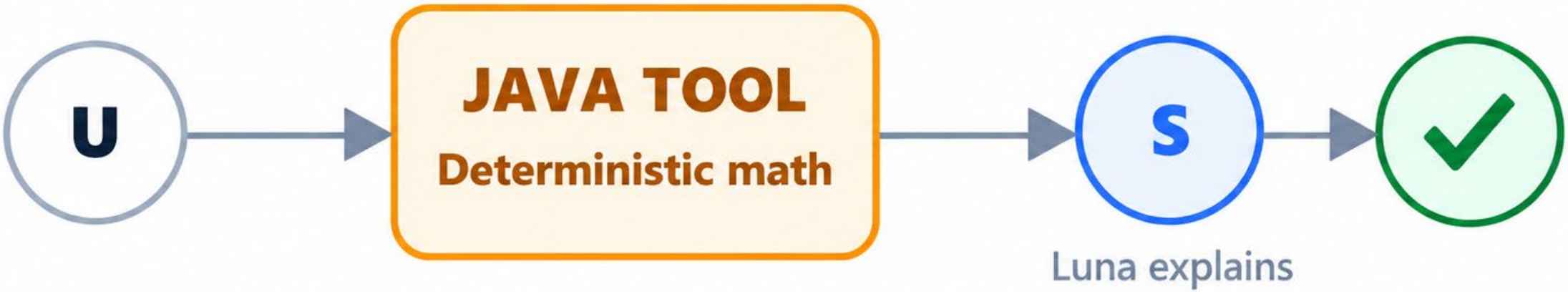



2000
1500
1000
500
0
tokens
1
2
3
4
5
6
7
8
9
10
11
12
conversation turn
Without compaction
With compaction

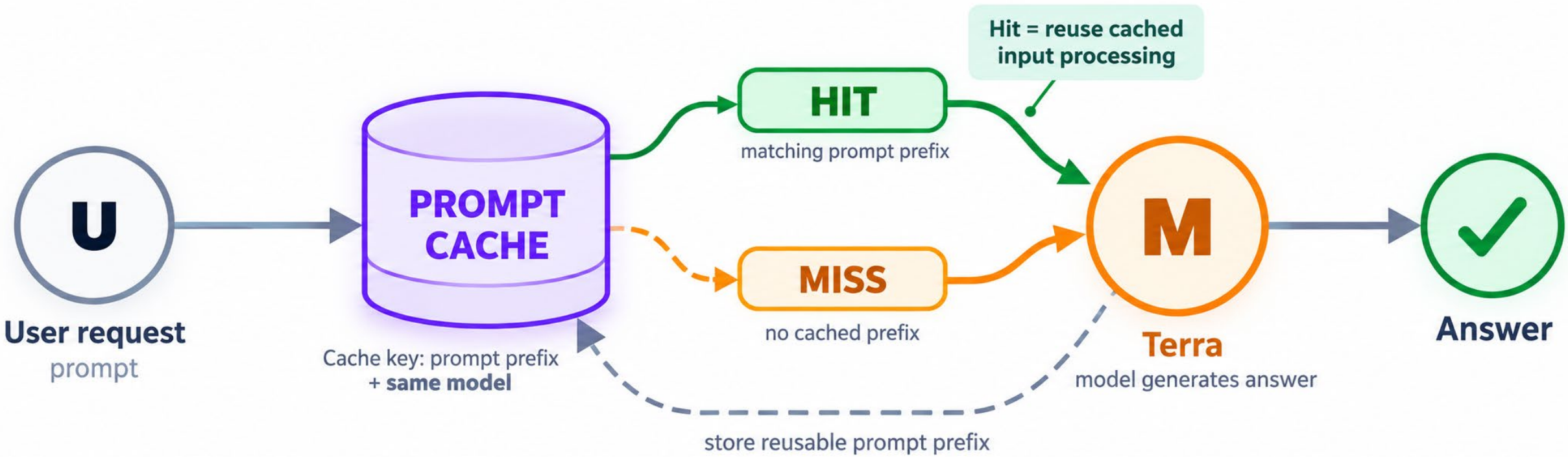
Retrieval-Augmented Generation (RAG)



Tools

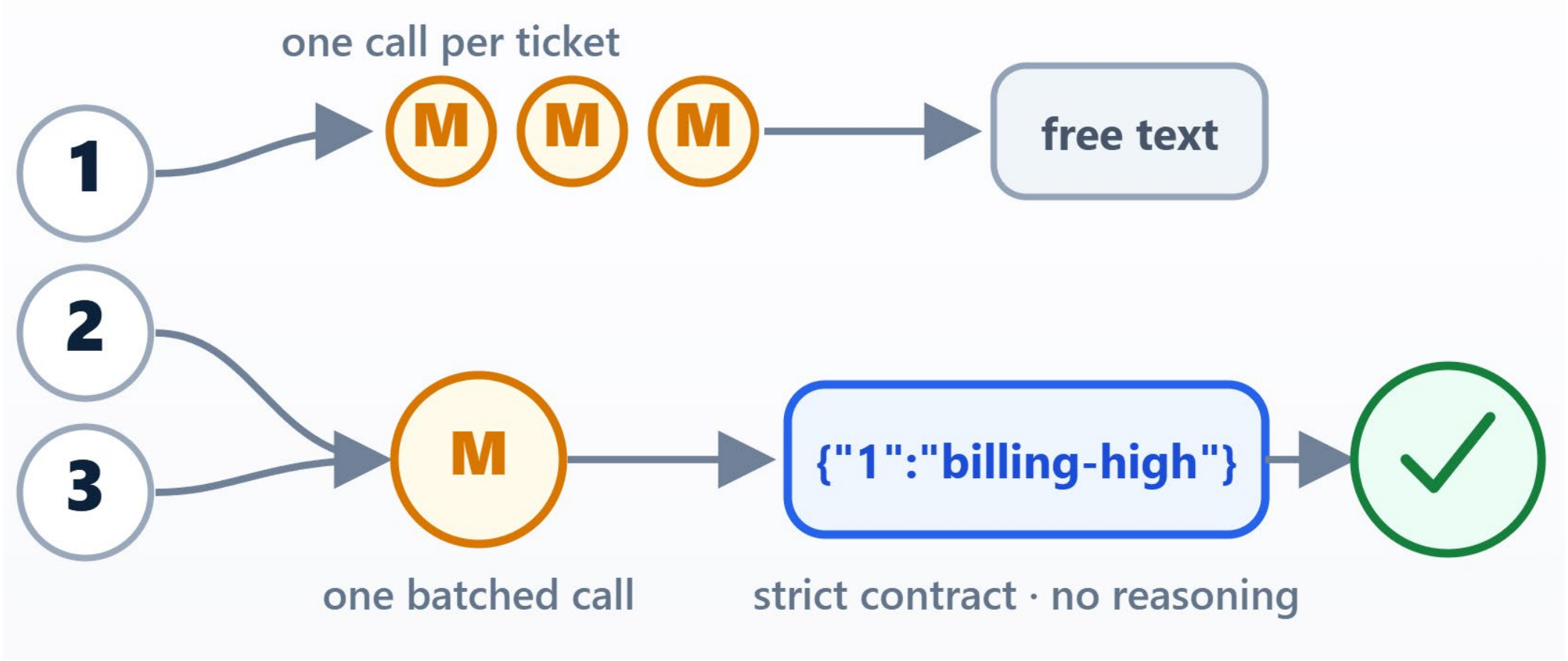


Caching


Hit = reuse cached input processing
HIT
matching prompt prefix
PROMPT CACHE
U
User request
prompt
MISS
no cached prefix
M
Terra
model generates answer
Answer
Cache key: prompt prefix + same model
store reusable prompt prefix


MODEL
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RAG
Tool use
Caching
Batching

Batching


one call per ticket
1
M
M
M
free text
2
3
M
{"1":"billing-high"}
one batched call
strict contract · no reasoning


MODEL
PROMPT
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OUTPUT
Router
Triage
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Batching

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= 4 tokens

Your sentence

Seven shapes make words into falling blocks.

8 tokens / o200k_base 44/500


Seven


␣shapes


␣make


␣words


␣into


␣falling

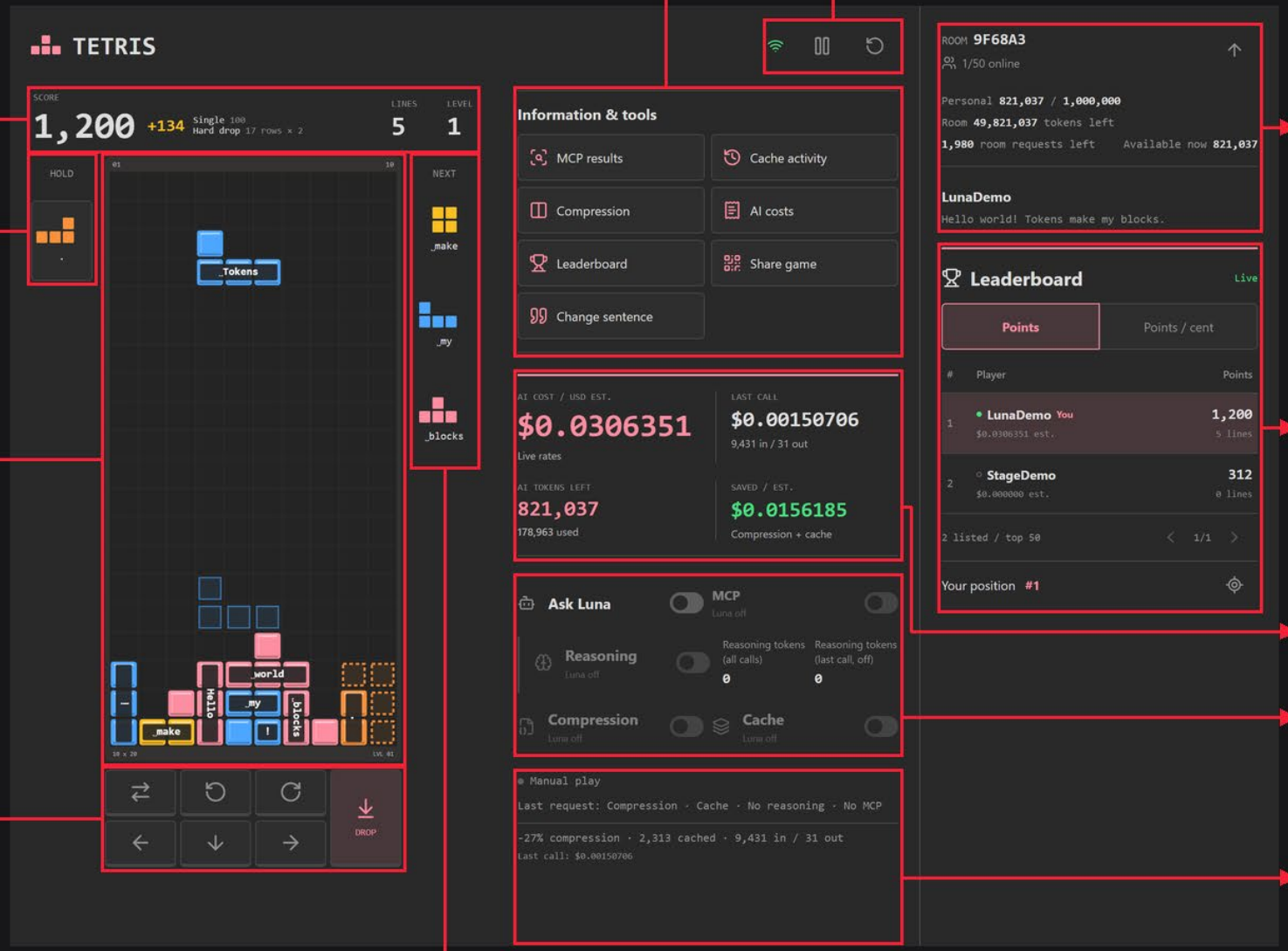

␣blocks


.

Join game


{
"BillingCurrency": "USD",
"CustomerEntityId": "Default",
"CustomerEntityType": "Retail",
"Items": [
{
"currencyCode": "USD",
"tierMinimumUnits": 0,
"retailPrice": 1.2,
"unitPrice": 1.2,
"armRegionName": "eastus",
"location": "US East",
"effectiveStartDate": "2026-08-01T00:00:00Z",
"meterId": "c6713056-7cef-508c-9e2e-7035984a4f44",
"meterName": "5.6 luna ShortCo Opt Std Gl 1M Tokens",
"productId": "DZH318Z0T9WD",
"skuId": "DZH318Z0T9WD/1166",
"productName": "Azure OpenAI GPT5",
"skuName": "5.6 luna ShortCo Opt Std Gl",
"serviceName": "Foundry Models",
"serviceId": "DZH31600VF1F",
"serviceFamily": "AI + Machine Learning",
"unitOfMeasure": "1M",
"type": "Consumption",
"isPrimaryMeterRegion": true,
"armSkuName": "56 luna ShortCo Opt Std Gl"
}
],
"NextPageLink": null,
"Count": 1
}
Currency: billing currency.
Retail price: list price charged.
Unit price: price per billing unit.
Region code: Azure region identifier.
Region name: human-friendly region name.
Effective date: when this price became active.
Meter name: full billing meter description.
Product name: product family.
SKU name: specific model/tier variant.
Service name: Azure service category.
Unit of measure: billed per 1M tokens.
Type: consumption-based billing.

TETRIS + LUNA



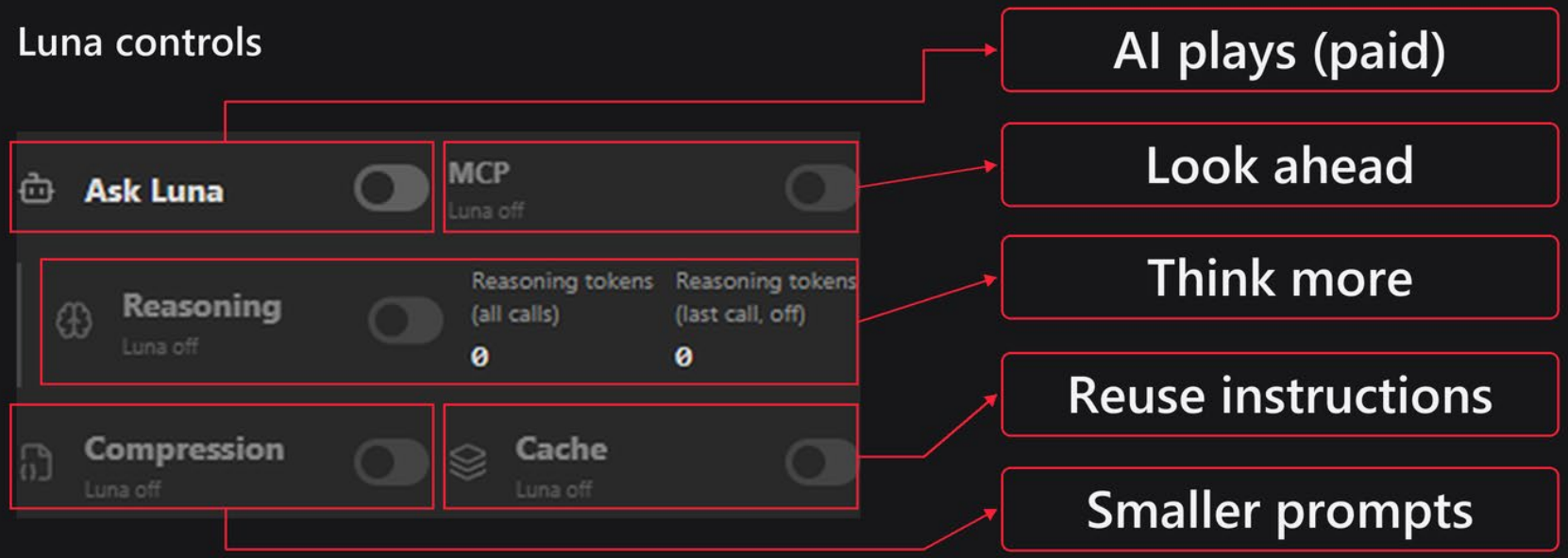
1. Pause / Restart
2. Score + points
3. Hold
4. Board
5. Next
6. Controls
7. Tools
8. AI cost
9. AI play
10. Status
11. Room
12. Rankings

AI COSTS + LUNA CONTROLS

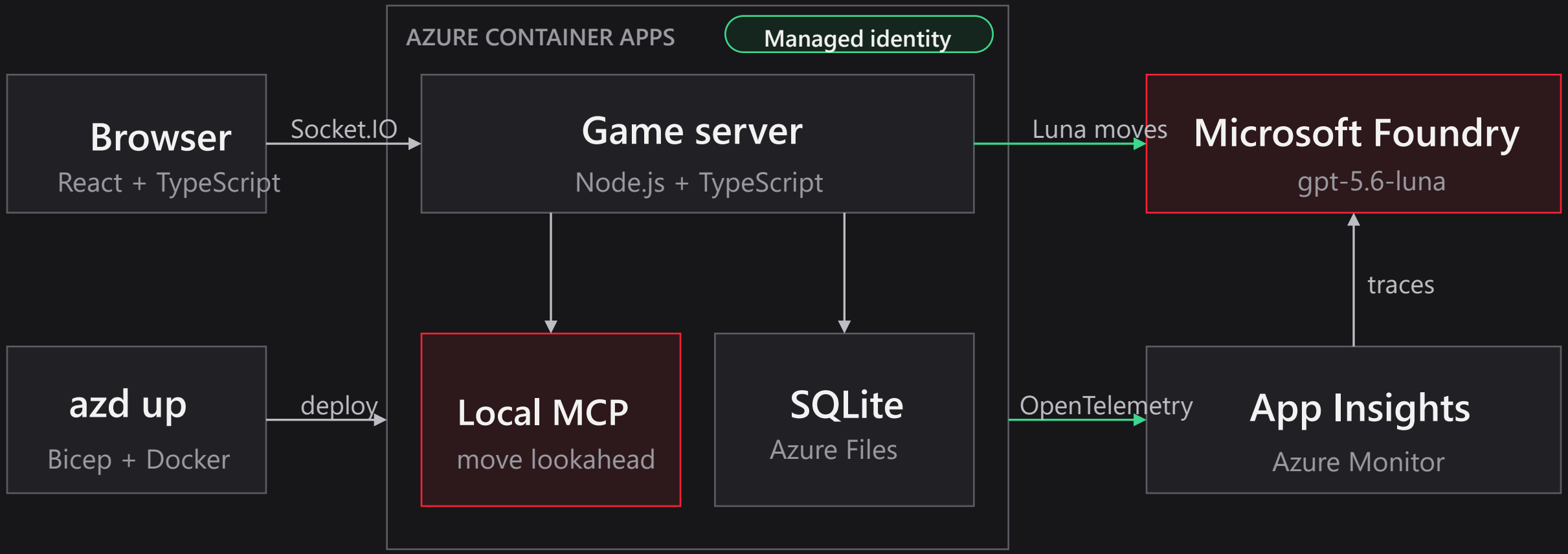
AI costs (USD estimates)

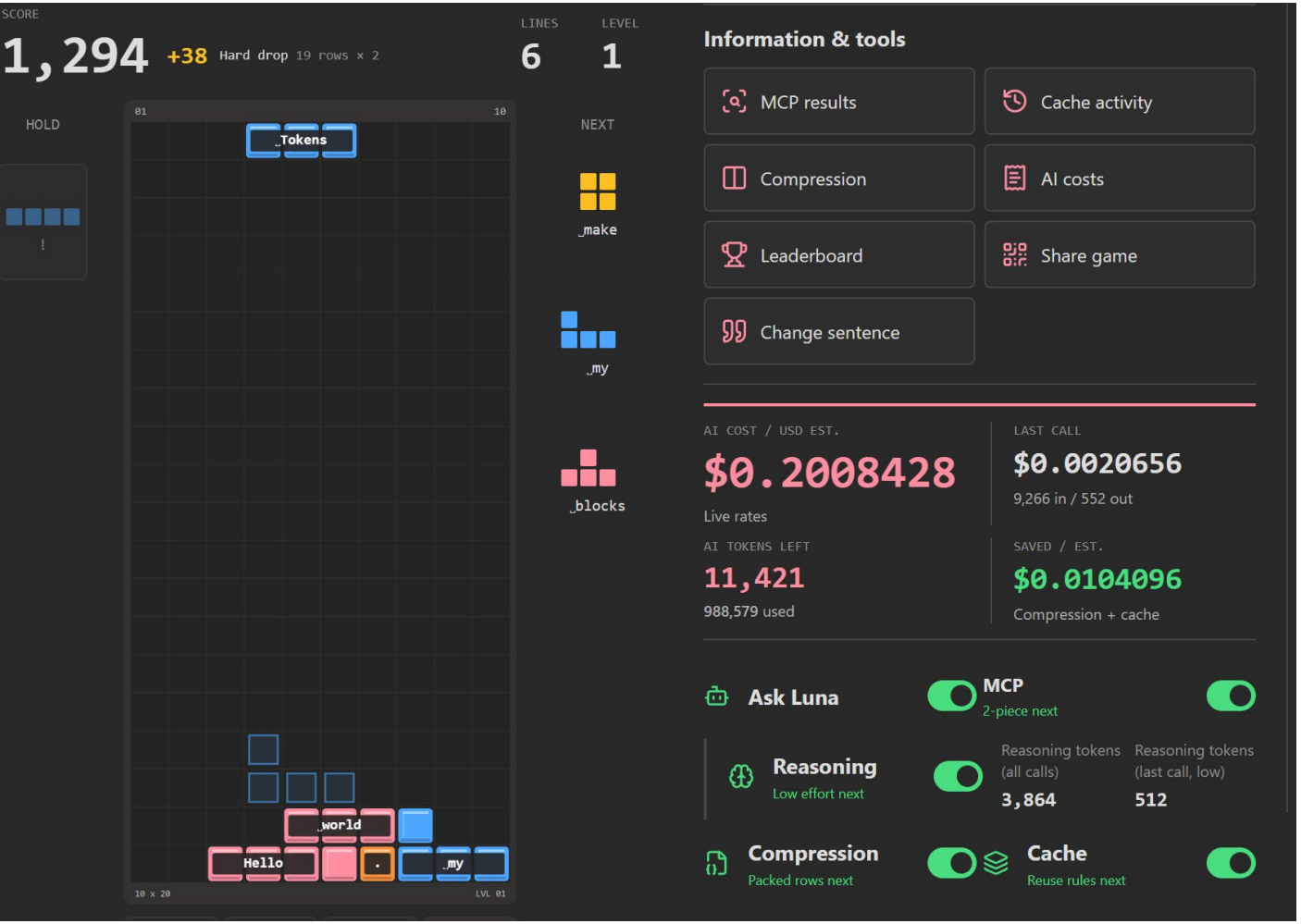


Luna controls



Architecture



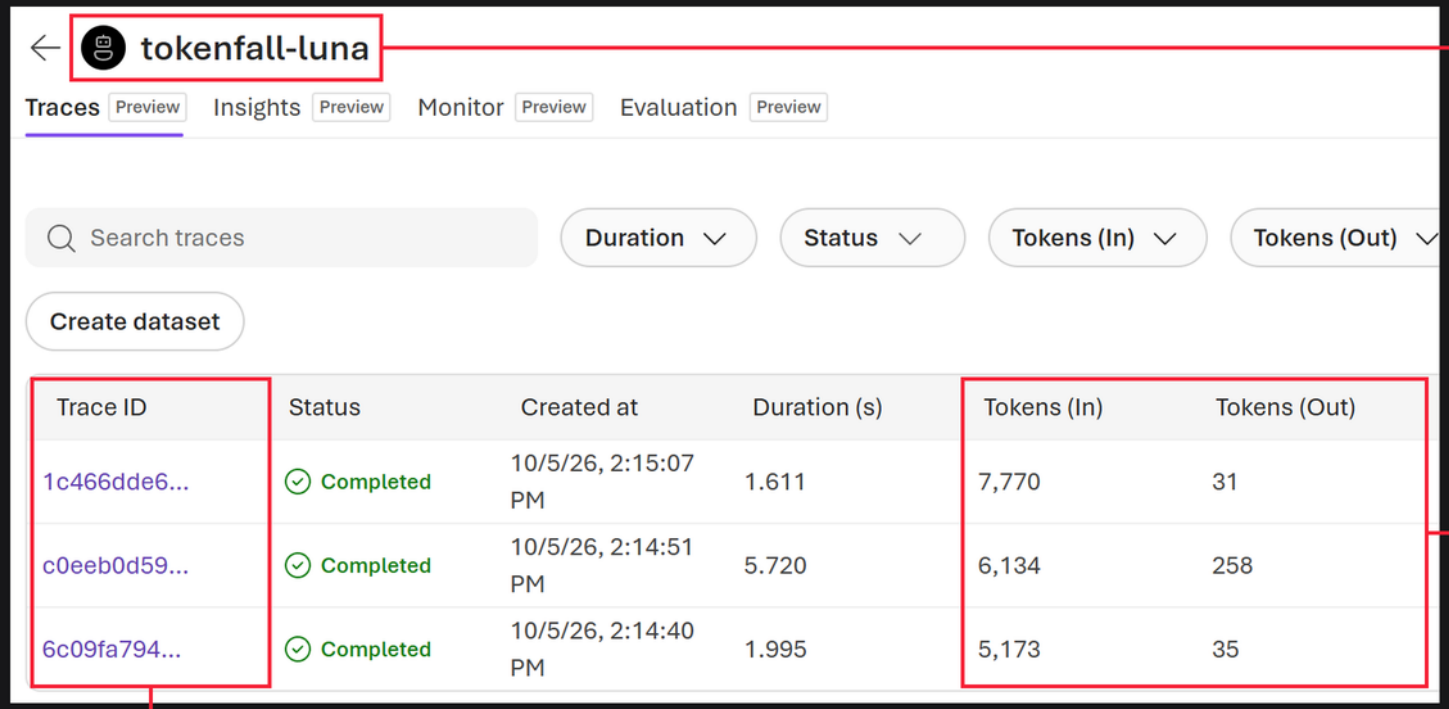


https://aka.ms/tokenfall

Agenda

- Introduction
- Demo 1 - 4 levers
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Tracing

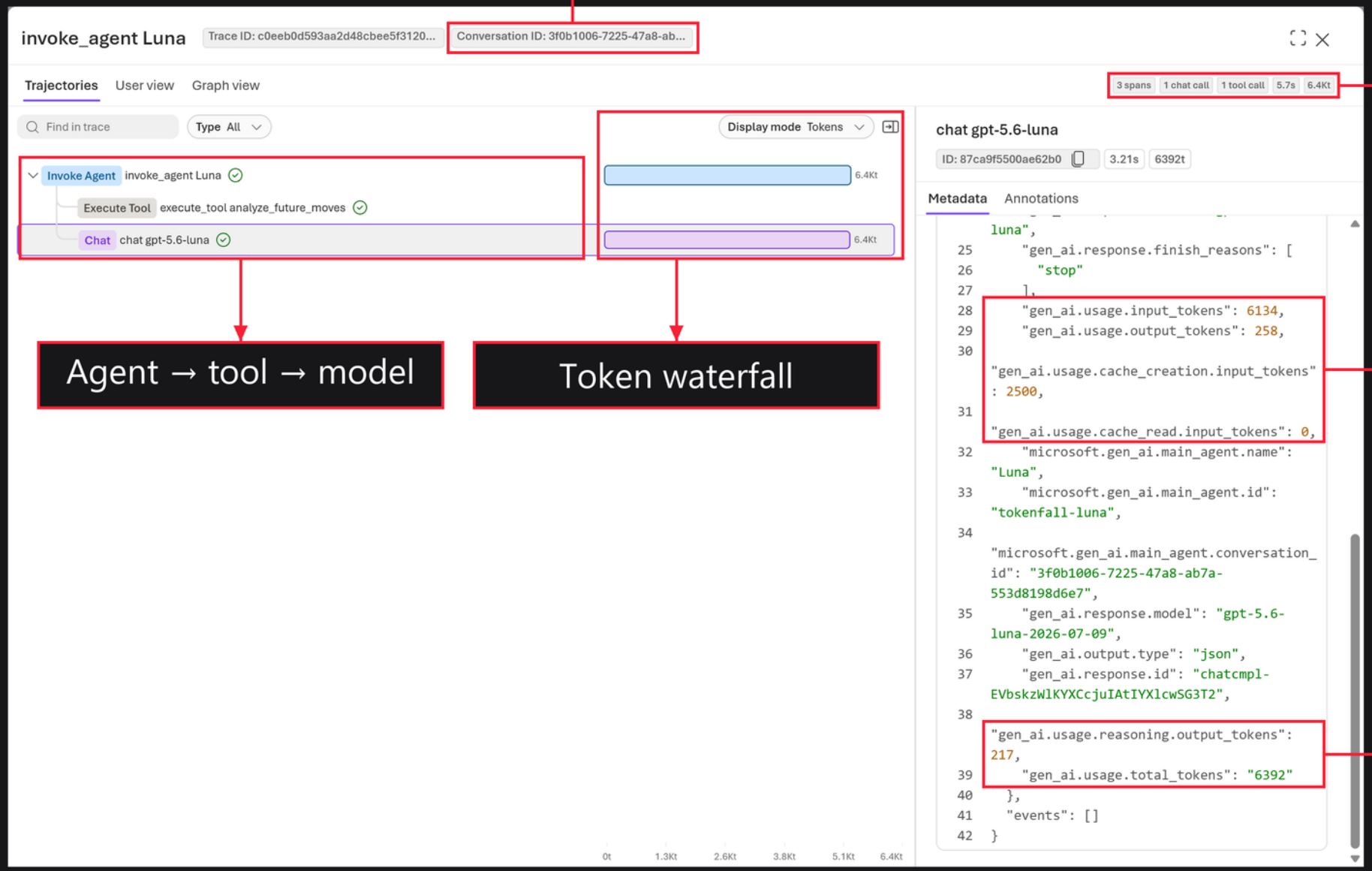


Linked Luna agent

Tokens in / out

One trace per move

One Luna move: MCP lookahead + reasoning



Agent → tool → model

Token waterfall

Game run

Trace totals

Input, output, cache

Reasoning + total

COST OF ONE MOVE WITH COMPRESSION AND CACHE

Measured in the game: the average of 31 Luna moves per option, all playing the same pieces

	UNCACHED INPUT	PER 1M INPUT	CACHED INPUT	PER 1M CACHED	OUTPUT TOKENS	PER 1M OUTPUT	PRICES ARE PER 1M TOKENS	PER MOVE
Plain All options off	(9,135	× $0.20 +	0	× $0.02 +	33	× $1.20)	÷ 1,000,000 ≈	**$0.00187**
Compression Board packed into rows	(6,904	× $0.20 +	0	× $0.02 +	33	× $1.20)	÷ 1,000,000 ≈	**$0.00142**
Cache Rules read from cache	(7,002	× $0.20 +	2,313	× $0.02 +	33	× $1.20)	÷ 1,000,000 ≈	**$0.00149**
Both Compression and cache	(4,654	× $0.20 +	2,313	× $0.02 +	32	× $1.20)	÷ 1,000,000 ≈	**$0.00102**

TOTAL FOR 31 MOVES, AS THE GAME'S AI COSTS PANEL SHOWED



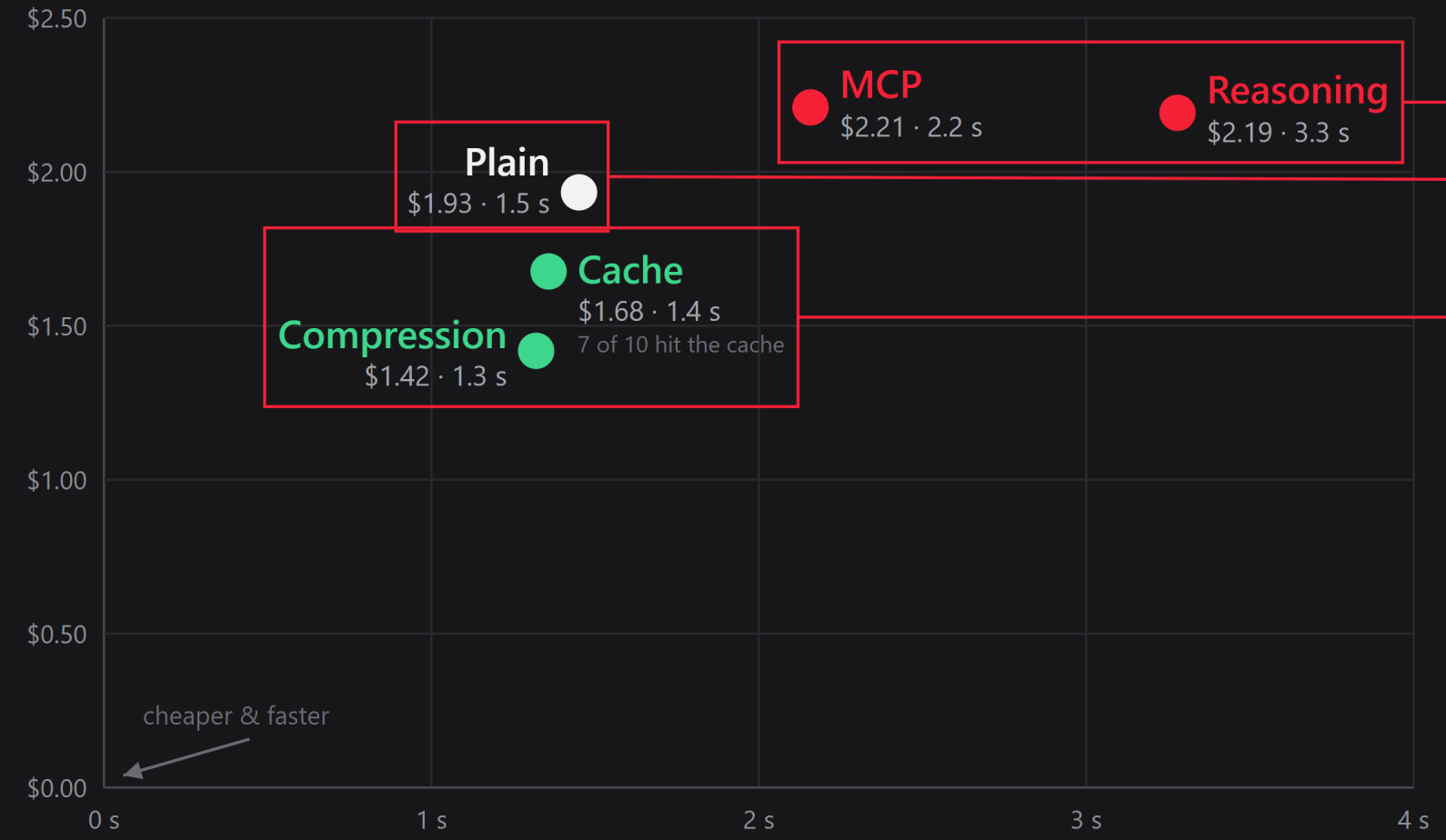
Cache rows show a cache hit. The first Cache move writes the 2,313 rule tokens at $0.25 instead: +$0.00012, once.

Compression and Cache together cut the cost by 45%

gpt-5.6-luna · Global Standard · Azure retail prices · played October 18, 2026

LUNA COST VS SPEED

Same 10 board positions for every option · game server with 2 CPUs



1 move ≈ $0.002

Costs more, 1.5–2.3× slower

Baseline: no options

Cheaper, same speed

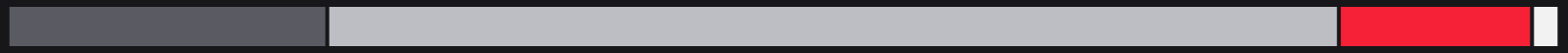
Foundry trace duration

Microsoft Foundry traces · gpt-5.6-luna · Azure retail prices

Tokens × Azure price

WHERE MCP'S $2.21 GOES

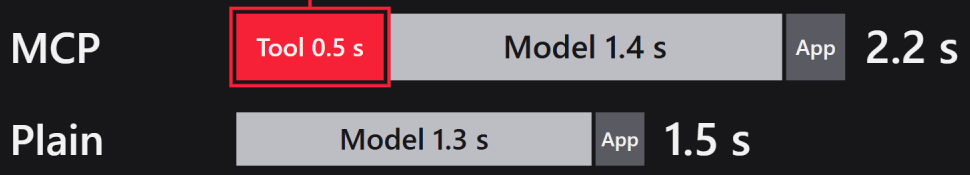
Cost of 1,000 Luna moves with MCP, averaged from 10 Foundry traces



	TOKENS PER MOVE	× PRICE PER 1M	= PER 1,000 MOVES	
Instructions Game rules and reply format	2,276	$0.20	**$0.46**	Same every move: cacheable
Game state Board, pieces and every legal move	7,200	$0.20	**$1.44**	Most of the cost: 65%
MCP lookahead Tool results for the next two moves	1,375	$0.20	**$0.27**	What MCP adds: +14%
Luna's reply The chosen move and a short tip	33	$1.20	**$0.04**	Output: 6× price per token
Total	10,884		**$2.21**	= $0.0022 per move

Tool: +0.5 s, no AI call
was 2.1 s on 0.5 CPU

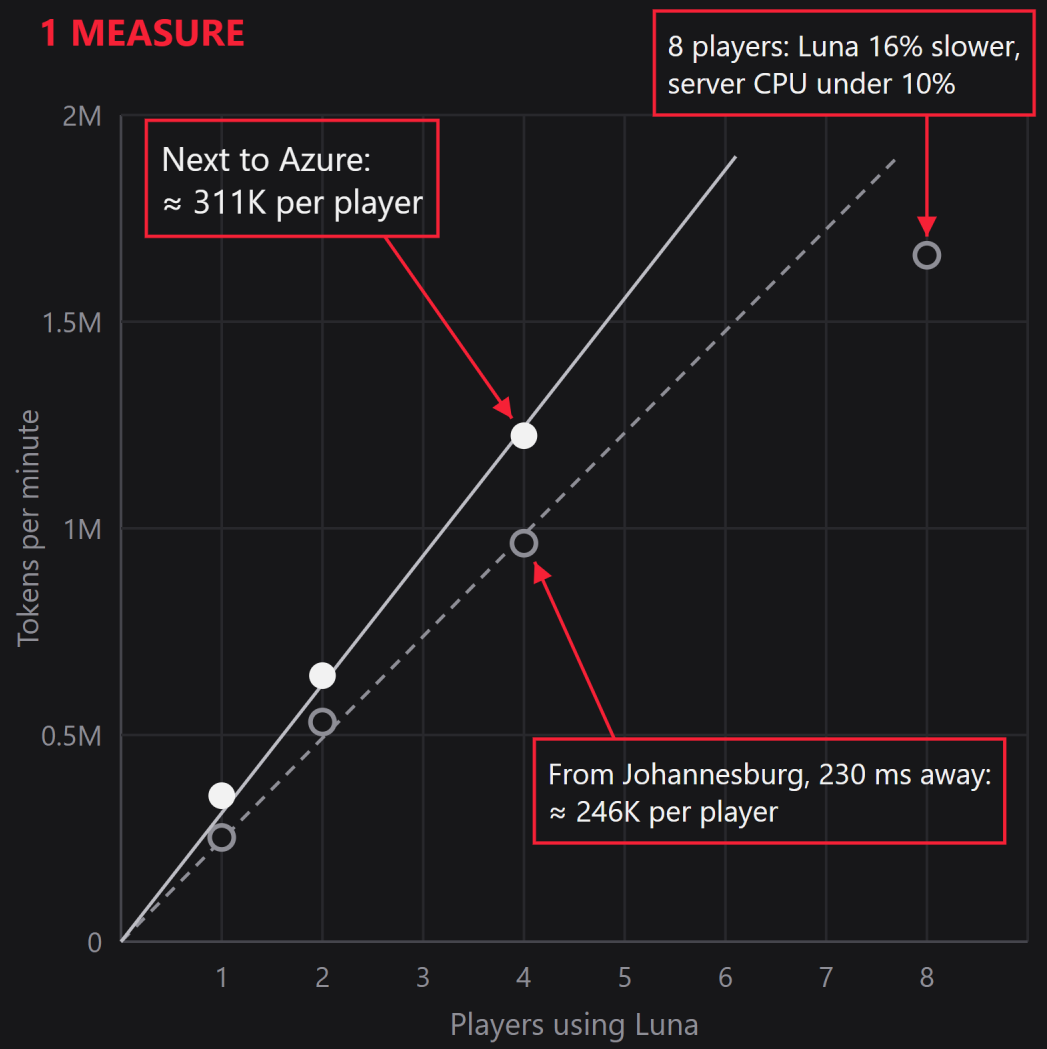
TIME PER MOVE



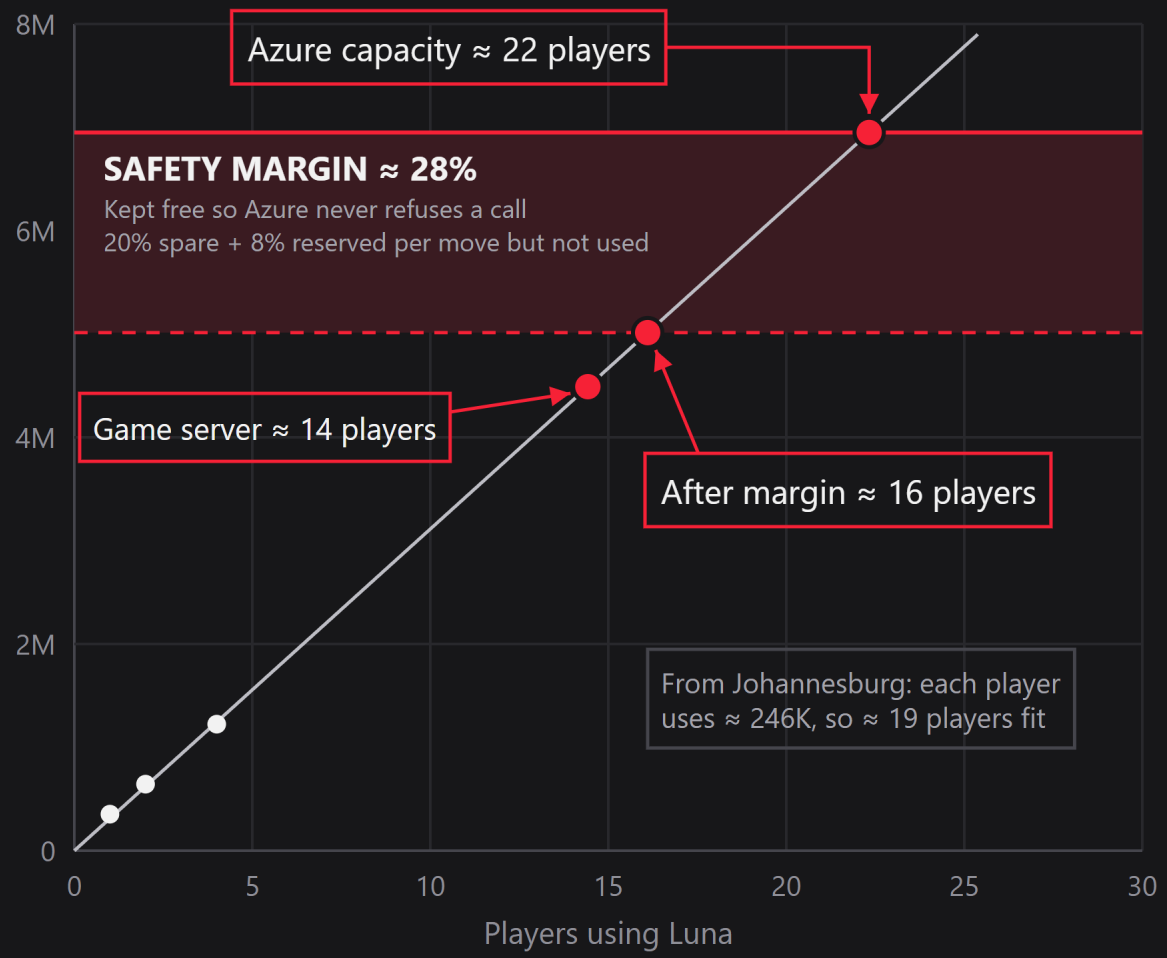
HOW MANY LUNA PLAYERS AT ONCE?

Plain Luna moves on the live game server: 1 replica, 2 CPUs, 4 GiB

1 MEASURE



2 PREDICT



Plain moves, one minute per step · Next to Azure: bots inside the app's container · Johannesburg: bots on the internet, 230 ms away

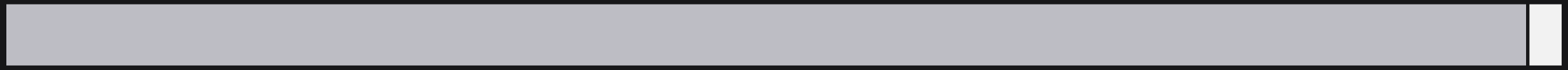
COST OF ONE MOVE: LUNA, TERRA AND SOL

The same plain move, with no cache: about 10,000 input and 35 output tokens at each model's prices

	INPUT TOKENS	PER 1M INPUT	OUTPUT TOKENS	PER 1M OUTPUT	PRICES ARE PER 1M TOKENS	PER MOVE
Luna gpt-5.6-luna	(10,000 ×	$0.20 +	35 ×	$1.20) ÷	1,000,000 ≈	**$0.002**
Terra gpt-5.6-terra	(10,000 ×	$2.00 +	35 ×	$12.00) ÷	1,000,000 ≈	**$0.020**
Sol gpt-5.6-sol	(10,000 ×	$4.00 +	35 ×	$20.00) ÷	1,000,000 ≈	**$0.041**

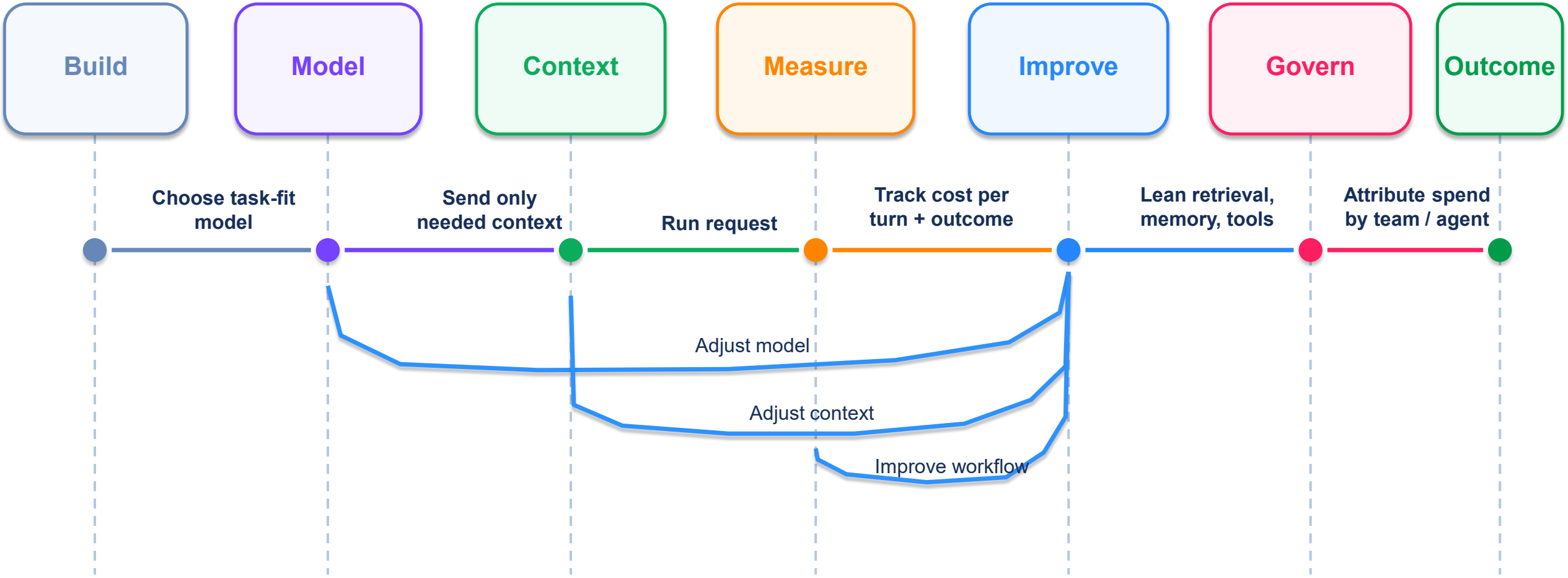
WHERE THE COST GOES

INPUT $0.002000 OUTPUT $0.000042



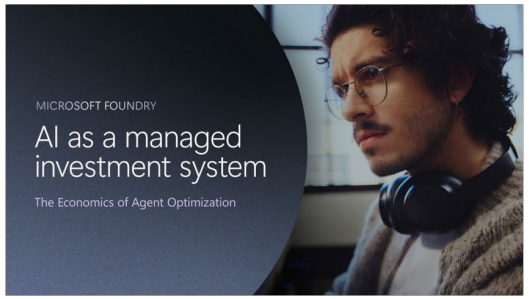
Input is about 98% of that

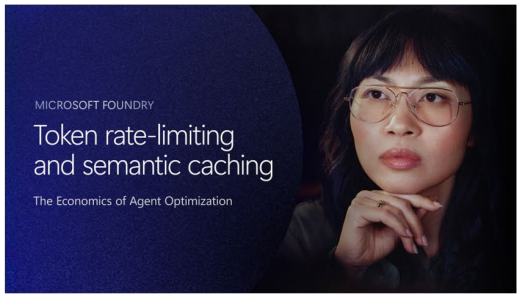
Token Economics Checklist

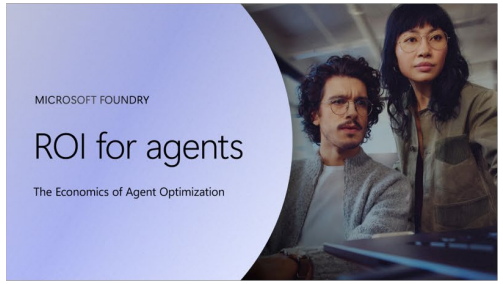


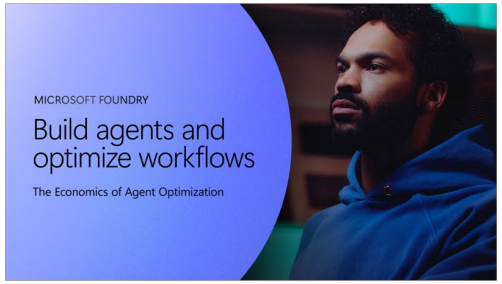
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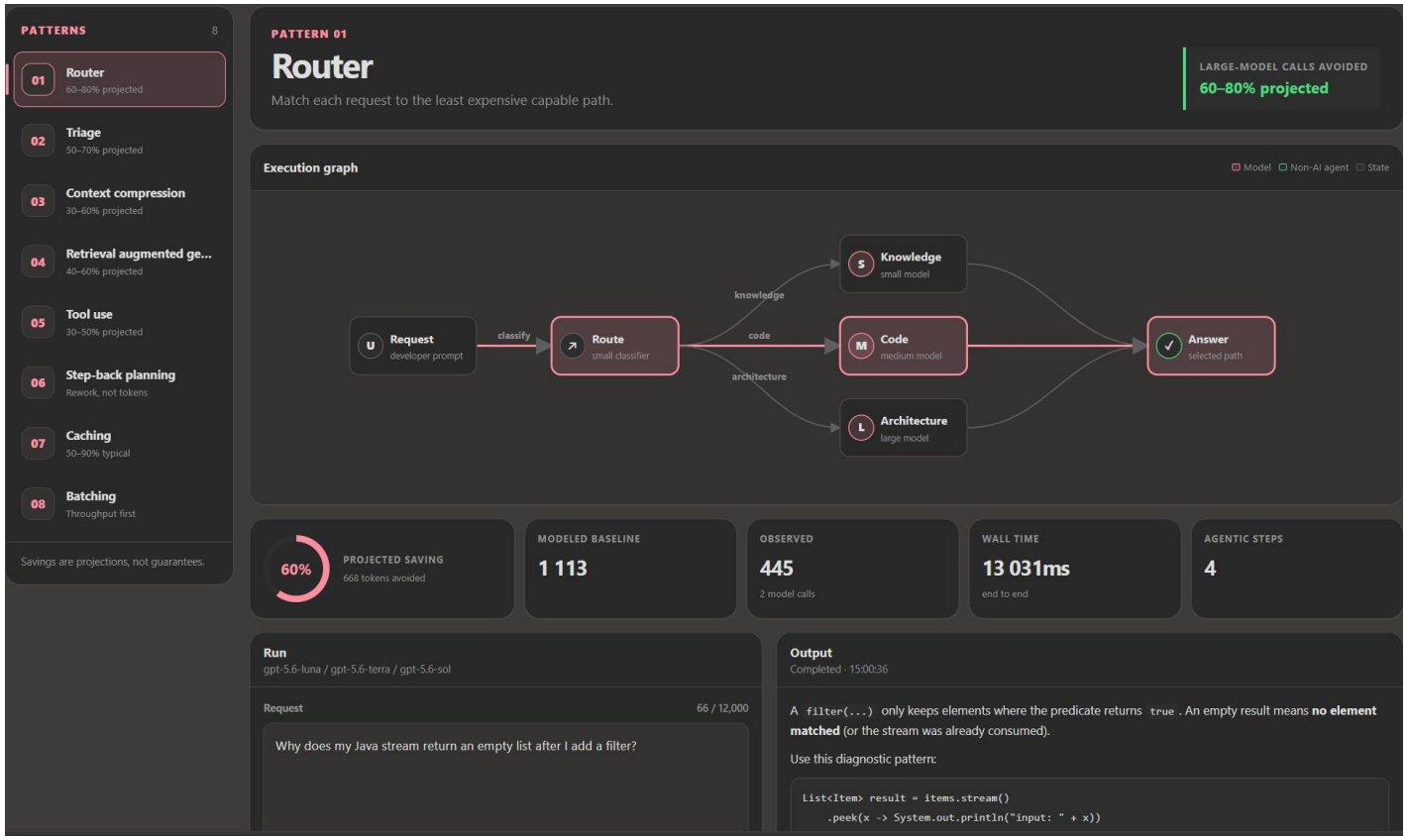




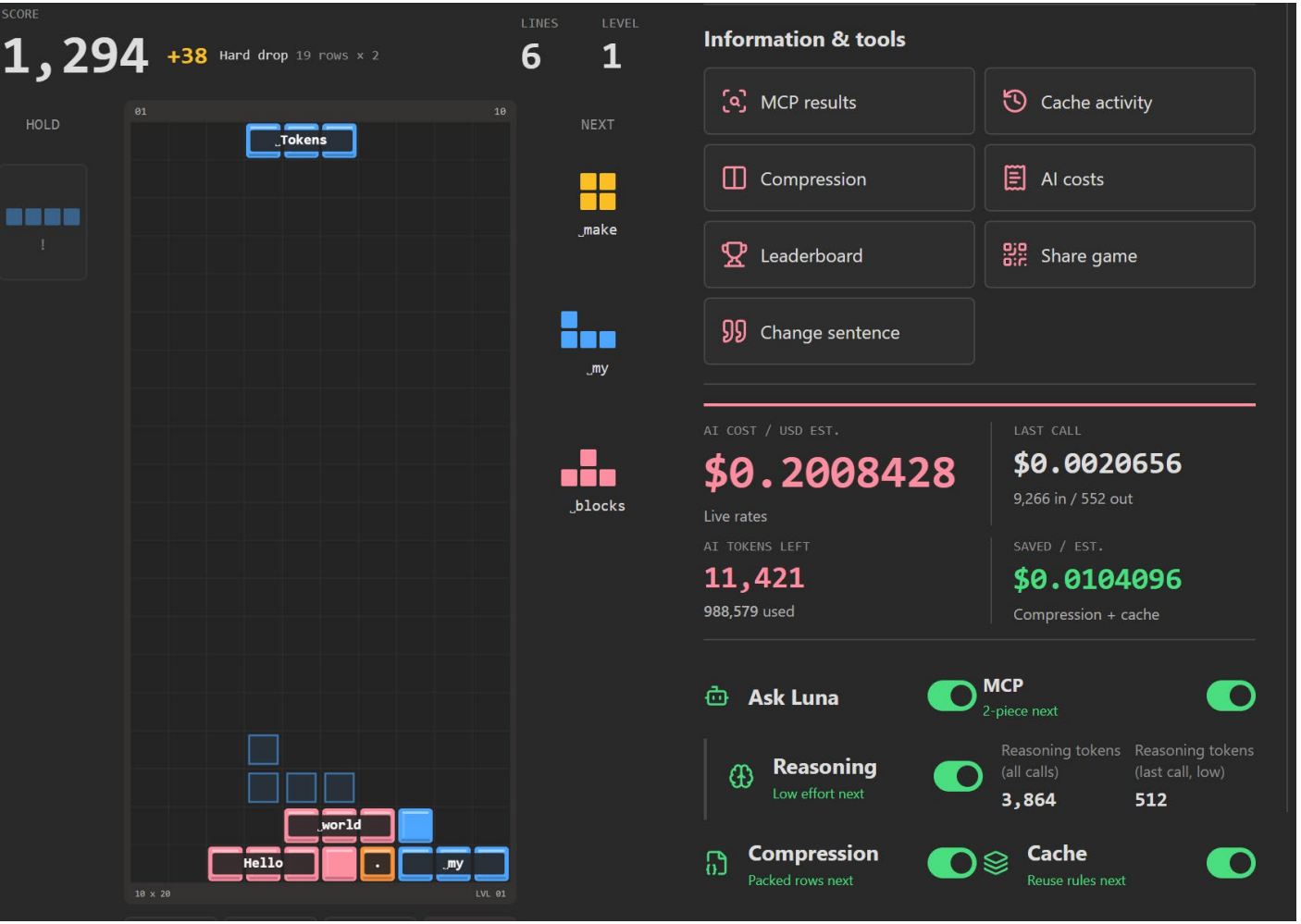




https://aka.ms/token-economics



https://github.com/roryp/token-design-patterns



https://github.com/roryp/tokenfall